QUOT OVER B X P^N_Z WITH THE PULLED-BACK O(1) IS PROPER

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1192-1201 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 1203-1230, the complete author proof (26 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 15 of queue_moduli.json (pb=1099, stmt 1192-1201, proof 1203-1230).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 01MW: sources/stacks-moduli/pinned-original-source.tex, source lines 1230–1230. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DPA: sources/stacks-moduli/pinned-original-source.tex, source lines 1047–1056. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DP5: sources/stacks-moduli/pinned-original-source.tex, source lines 925–930. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DP7: sources/stacks-moduli/pinned-original-source.tex, source lines 984–999. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01PS: sources/stacks-moduli/pinned-original-source.tex, source lines 1230–1230. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01Q3: sources/stacks-moduli/pinned-original-source.tex, source lines 1230–1230. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 083R: sources/stacks-moduli/pinned-original-source.tex, source lines 1230–1230. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let B be an algebraic space. Let $X = B \times \mathbf{P}^n_{\mathbf{Z}}$. Let $\mathcal{L}$ be the pullback of $\mathcal{O}_{\mathbf{P}^n}(1)$ to X. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module of finite presentation. The algebraic space $\mathrm{Quot}^P_{\mathcal{F}/X/B}$ parametrizing quotients of $\mathcal{F}$ having Hilbert polynomial P with respect to $\mathcal{L}$ is proper over B.*

Proof. The question is étale local over B, see Morphisms of Spaces, Lemma 083R. Thus we may assume B is an affine scheme. In this case $\mathcal{L}$ is an ample invertible module on X (by Constructions, Lemma 01MW and the definition of ample invertible modules in Properties, Definition 01PS). Thus we can find $r' \geq 0$ and $r \geq 0$ and a surjection

$$\mathcal{O}_X^{\oplus r} \longrightarrow \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes r'}$$

by Properties, Proposition 01Q3. By Lemma 0DP7 we may replace $\mathcal{F}$ by $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes r'}$ and $P(t)$ by $P(t + r')$. By Lemma 0DP5 we obtain a closed immersion

$$\mathrm{Quot}^P_{\mathcal{F}/X/B} \longrightarrow \mathrm{Quot}^P_{\mathcal{O}_X^{\oplus r}/X/B}$$

Since we've shown that $\mathrm{Quot}^P_{\mathcal{O}_X^{\oplus r}/X/B} \to B$ is proper in Lemma 0DPA we conclude. □